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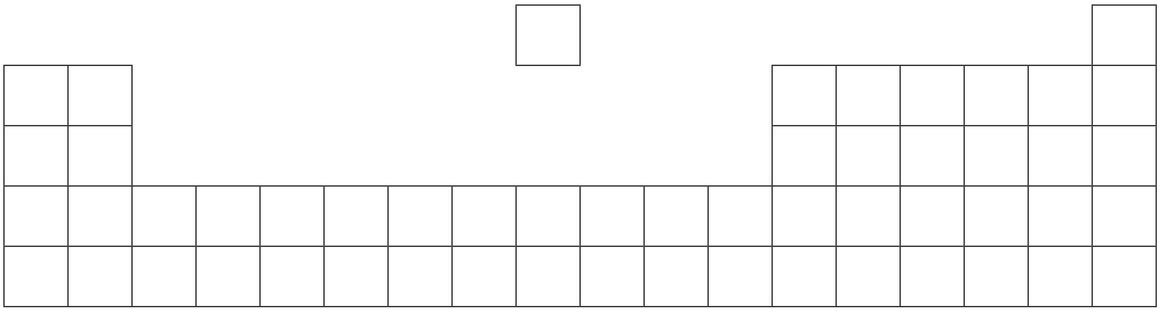
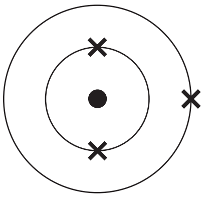
5. The following diagram is an outline of the Periodic Table.



(a) Write letters **A**, **B** and **C** on the **diagram** in the positions of the elements that fit the following descriptions. [3]

A the element with the electronic structure



B the element in Group 2 and Period 4

C an element that shows both metallic and non-metallic properties

(b) The following table shows the atomic numbers and boiling points of the inert gases.

Inert gas	helium	neon	argon	krypton	xenon
Atomic number	2	10	18	36	54
Boiling point (°C)	−269	−246	−186	−153	−108

(i) Plot this data on the grid opposite. Draw a suitable line. [3]

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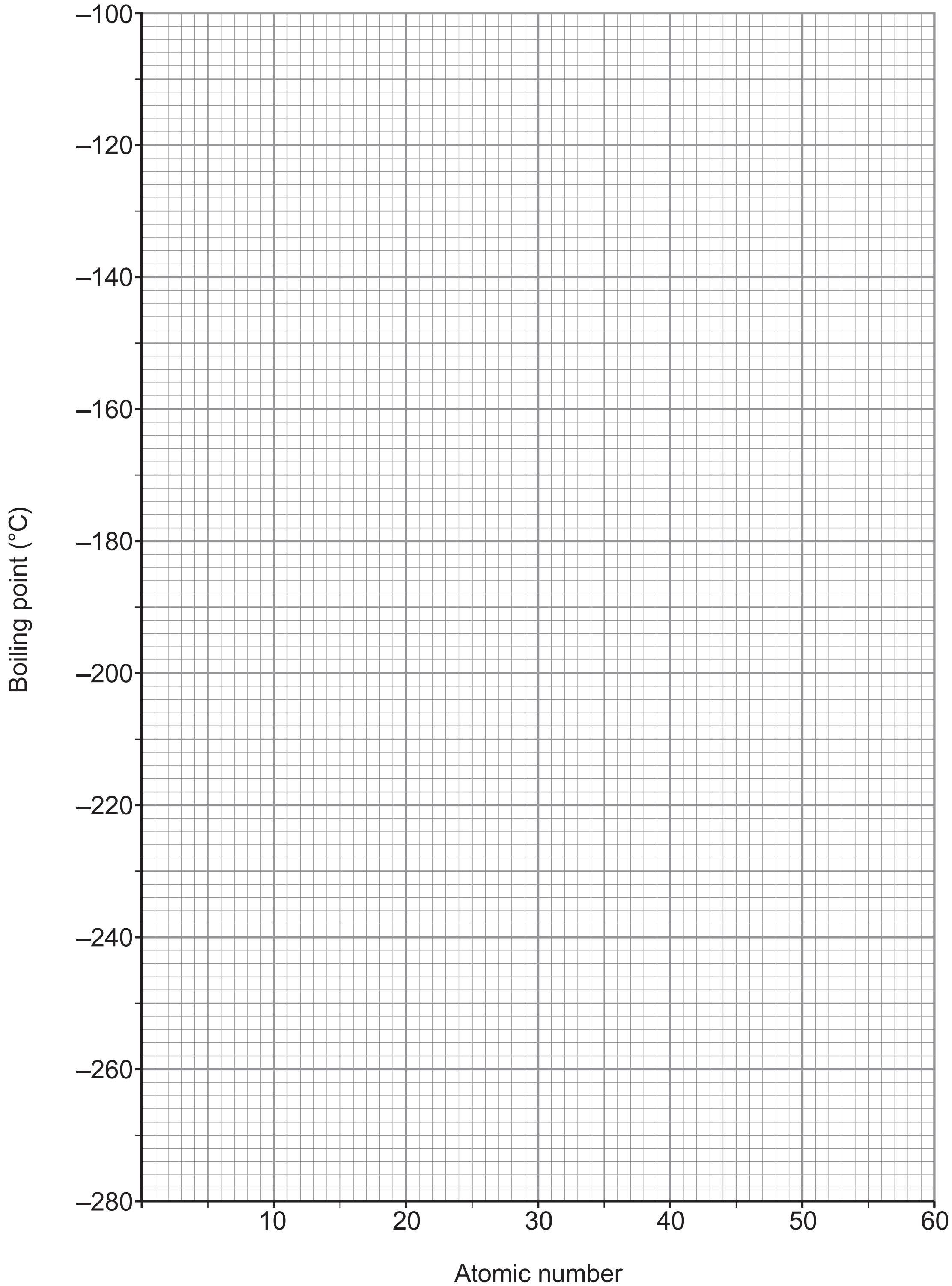

(ii) Describe the trend in boiling point shown on the graph. [1]

....

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(c) The following table shows the boiling points of the inert gases in both °C and K.

Inert gas	helium	neon	argon	krypton	xenon
Boiling point (°C)	−269	−246	−186	−153	−108
Boiling point (K)	4	27	....	120	165

Use the information in the table to calculate the boiling point of argon in K. [2]

Boiling point = ... K

(d) Give **one** use of argon. Explain in terms of electronic structure why it is used for this purpose. [2]

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